Developer: Júlia Stella Silva de Paula

Development Approach

Due to last-minute unforeseen circumstances, my available time to complete the project was reduced from 24 hours to a maximum of 12 hours. To overcome this constraint, I decided to structure the project into a single phase and adopted a delegation strategy: I assigned the PO/PM (Product Owner/Product Manager) and validation roles to Gemini's NotebookLM, using it as an AI assistant.

It is worth mentioning that I have previously developed an AI assistant using OpenAI's model in an enterprise environment.

During development, the configuration and coding stages delivered the greatest time savings. The workflow in the notebook was as follows: I uploaded the test dataset and started with an initial prompt to define a viable MVP scope aligned with the established deadlines. This process involved precisely mapping out what needed to be developed and the rationale behind each decision.

Based on this analysis, we established our action plan.

Action Plan (12-Hour Sprint)

1. Setup and Data Preparation (Hours 1–2)

- Initialize the Git repository immediately and make the initial commit.
- Load the `.csv` files using Pandas and handle only critical missing values that could break product and price lookup functionality.

2. LLM and Tool Integration (Hours 3–7)

- Develop the *System Prompt* to define the store assistant's persona and behavior.
- Implement a hybrid architecture:
    - **Function Calling (SQL/Pandas Agent):** Allow the LLM to query the CSV datasets and answer questions such as "How much does product X cost?"
    - **Basic RAG:** Generate embeddings for `politicas_da_loja.pdf` to retrieve relevant information and answer questions about return policies, business hours, and other store procedures.

3. Interface and Testing (Hours 8–10)

- Integrate the agent's logic into a Streamlit interface.
- Simulate the 3–5 required scenarios, ensuring that at least one involves a more complex query combining information from the PDF policies with actual dataset records.

- Export the conversations in `.md`, `.txt`, or image format.

4. Documentation and Delivery (Hours 11–12)

- Write the `README.md`, explaining and justifying the technical decisions (e.g., why Streamlit, a free LLM, and Pandas were selected).
- Document known limitations, outline what could be improved with more than 12 hours of development time, and transparently describe how auxiliary AI tools were used throughout the workflow (which can be considered a positive aspect of the development process).
- Reply to the recruitment email with the public repository link before the deadline.

Next, I created a second prompt focused on code generation to speed up repetitive, manual tasks and clarify technical questions, as this was my first experience working with Google's AI agents. The official setup documentation and API key management guidelines, available in the Google AI Studio documentation, were essential resources for generating the code.

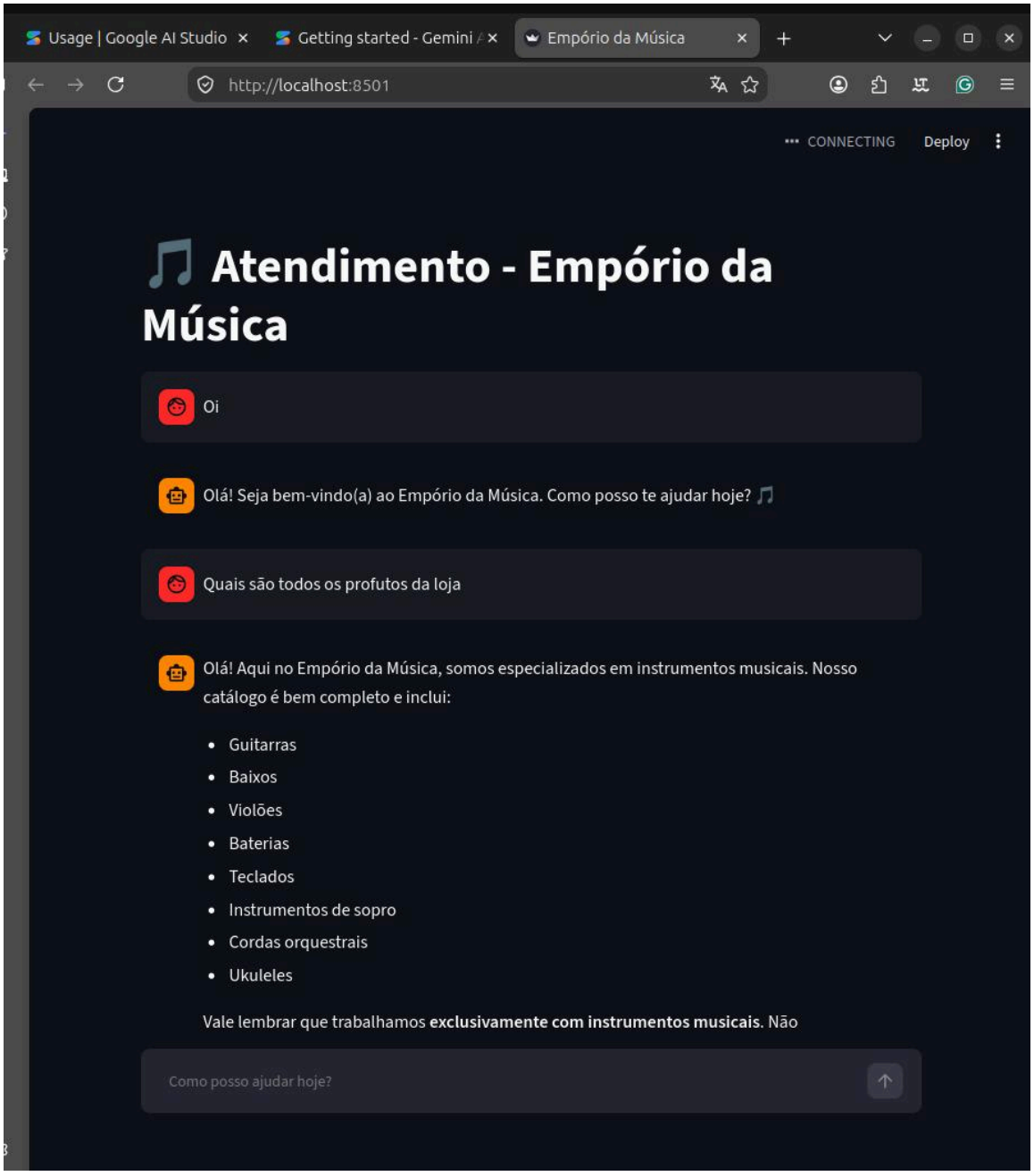
With this structure in place and the support of these prompts, I was able to streamline development and focus primarily on bug fixing and comprehensive testing. The project incorporated the following technical solutions:

- **Simplified RAG Approach:** Instead of setting up complex vector databases such as ChromaDB or FAISS, we extracted the full text from the store policy document (`politicas_da_loja.pdf`) using the PyPDF2 library and injected it directly into the Gemini model's `system_instruction`. This architectural decision takes advantage of the model's large context window, enabling immediate and accurate responses about exchange and return policies.
- **Function Calling for Tabular Data:** For product price and inventory queries, we avoided the overhead of complex ORMs. The `data_loader.py` script uses Pandas to read `products.csv` and handle critical missing values in the price and stock columns. The AI agent autonomously determines when to invoke the Python tool (`consultar_produto`) to retrieve real-time product data during execution.
- **Streaming Response Handling:** The interface was built using `send_message_stream` to provide a smooth user experience through incremental text generation. The response pipeline includes filters to ignore internal tool execution blocks (*Function Calling*), keeping the interface clean, responsive, and free from visual glitches.

Assistant Validation and Manual Testing

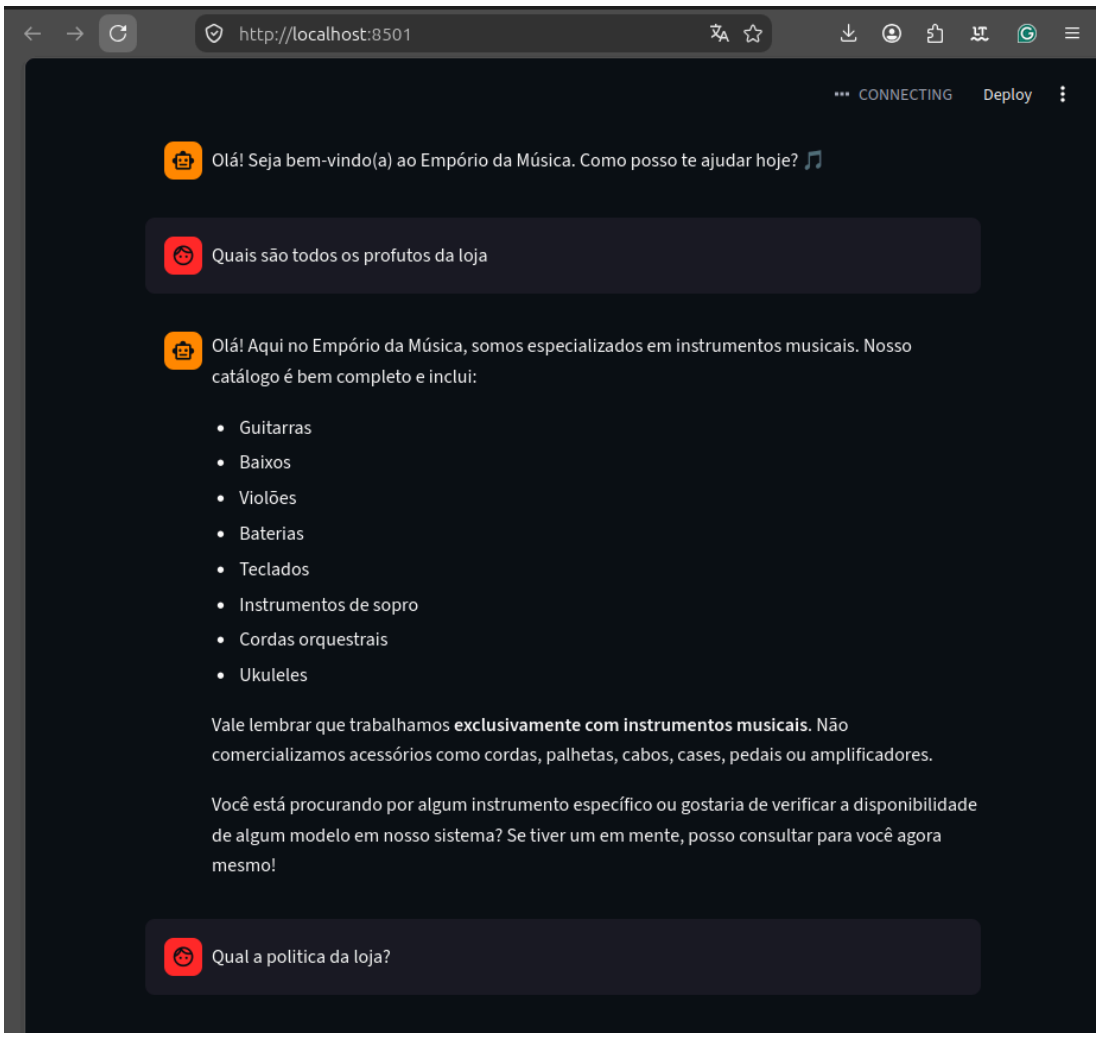
To ensure the AI agent responded accurately and consistently with the provided guidelines and knowledge base, I conducted a series of manual interactions simulating real customer service scenarios through the application interface:

1. **Conversation Start and Introduction:** The test began with a simple greeting ("*Hi*"). The assistant responded promptly and in a welcoming tone, introducing the business as "*Empório da Música*" and inviting the customer to ask for assistance.
1.

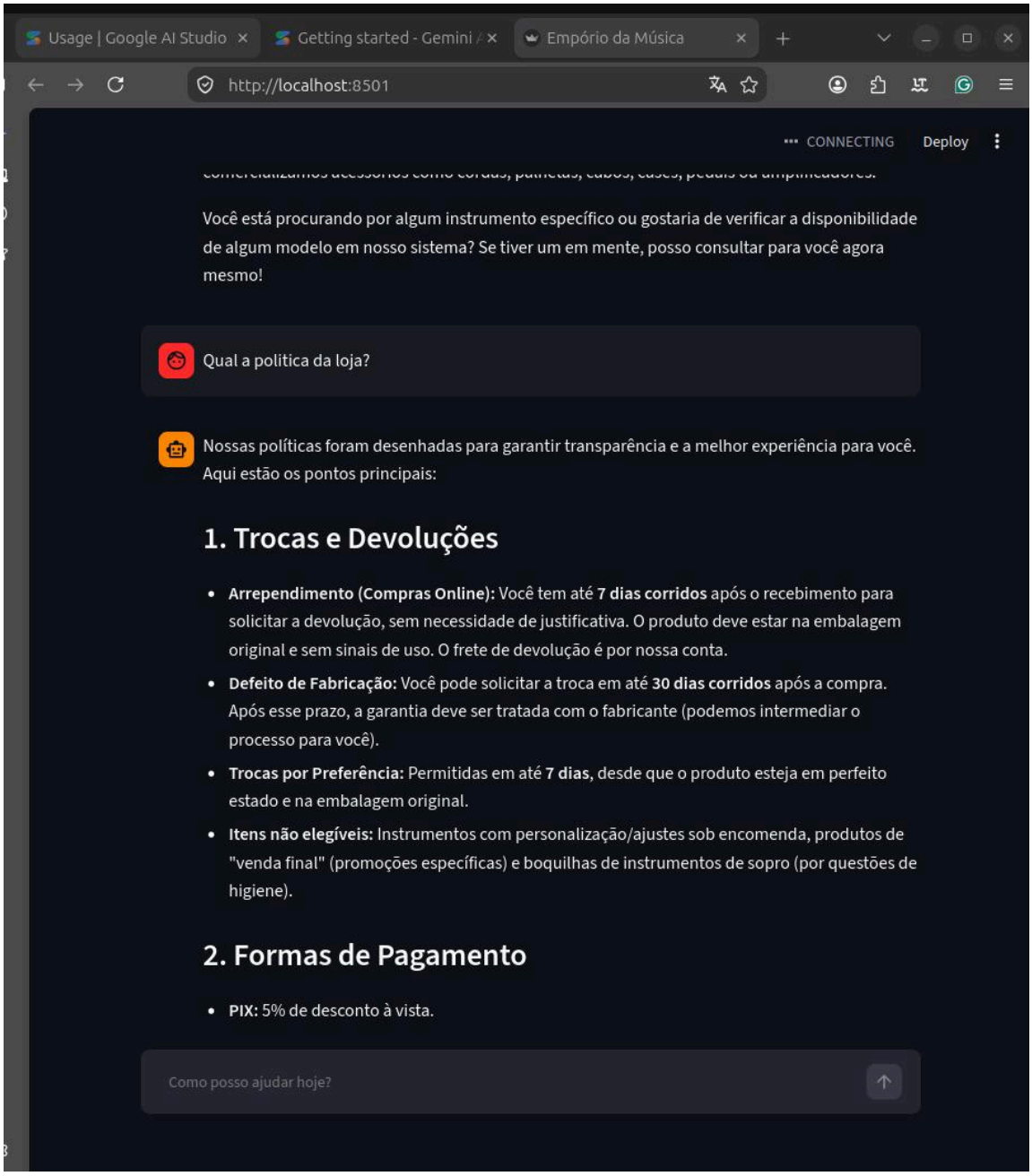


2. **Product Catalog Inquiry:** Next, I asked about the products available in the store ("What products do you have available?"). The agent accurately listed the store's musical instrument catalog, including guitars, bass guitars, acoustic guitars, drums, keyboards,

wind instruments, orchestral string instruments, and ukuleles, highlighting the business's exclusive focus on this product category.

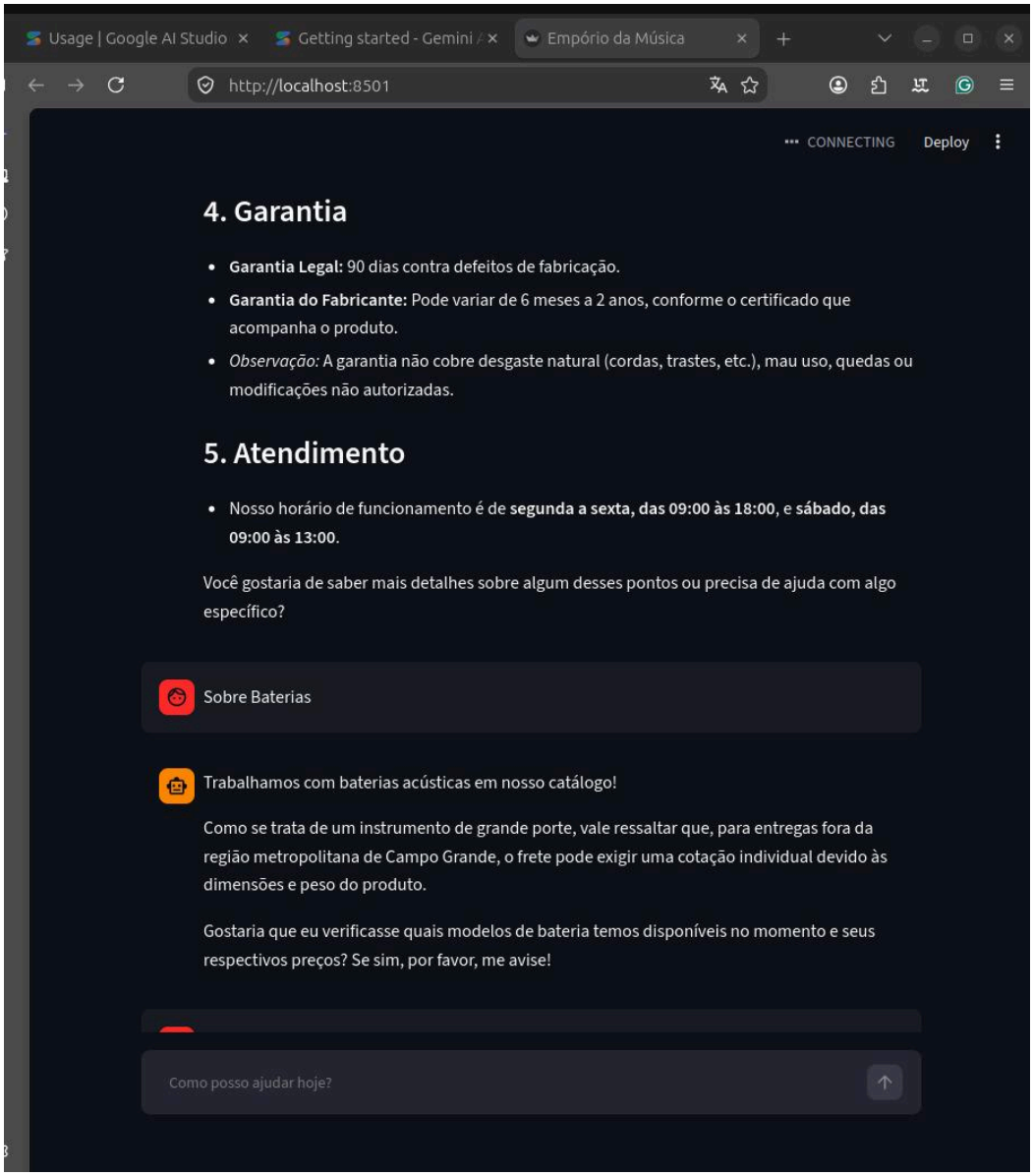


3. **Store Policy Verification:** The testing process then moved on to validating the store's internal guidelines using the question, "What is the store's policy?". The assistant successfully retrieved the contextual information provided through the RAG-based approach, using the store policy document as its knowledge source. It organized the response into clear sections: **Exchanges and Returns** (7-day return period for withdrawal and 30 days for defective products), **Payment Methods** (PIX, credit/debit cards, and bank slips), **Shipping and Delivery**, **Warranty**, and **Customer Support**.

4. 
Usage | Google AI Studio
Getting started - Gemini
Empório da Música
http://localhost:8501
CONNECTING
Deploy
Você está procurando por algum instrumento específico ou gostaria de verificar a disponibilidade de algum modelo em nosso sistema? Se tiver um em mente, posso consultar para você agora mesmo!
Qual a politica da loja?
Nossas políticas foram desenhadas para garantir transparência e a melhor experiência para você. Aqui estão os pontos principais:
1. Trocas e Devoluções
Arrependimento (Compras Online): Você tem até 7 dias corridos após o recebimento para solicitar a devolução, sem necessidade de justificativa. O produto deve estar na embalagem original e sem sinais de uso. O frete de devolução é por nossa conta.
Defeito de Fabricação: Você pode solicitar a troca em até 30 dias corridos após a compra. Após esse prazo, a garantia deve ser tratada com o fabricante (podemos intermediar o processo para você).
Trocas por Preferência: Permitidas em até 7 dias, desde que o produto esteja em perfeito estado e na embalagem original.
Itens não elegíveis: Instrumentos com personalização/ajustes sob encomenda, produtos de "venda final" (promoções específicas) e boquilhas de instrumentos de sopro (por questões de higiene).
2. Formas de Pagamento
PIX: 5% de desconto à vista.
Como posso ajudar hoje?



5. **In-Depth Inquiry About Specific Instruments:** Finally, I tested the agent's ability to handle category-specific inquiries by asking, "Tell me about drums." The agent correctly identified the product category, confirmed that the store carries acoustic drum kits, and provided an important logistical clarification regarding shipping fees and delivery quotes for locations outside the Campo Grande metropolitan area. This demonstrated strong contextual understanding and a proactive approach, as the assistant also offered to check product availability.



Manual testing confirmed that injecting the store policy document directly into the model worked as intended, delivering fast, well-structured responses while preventing hallucinations in business-related information.